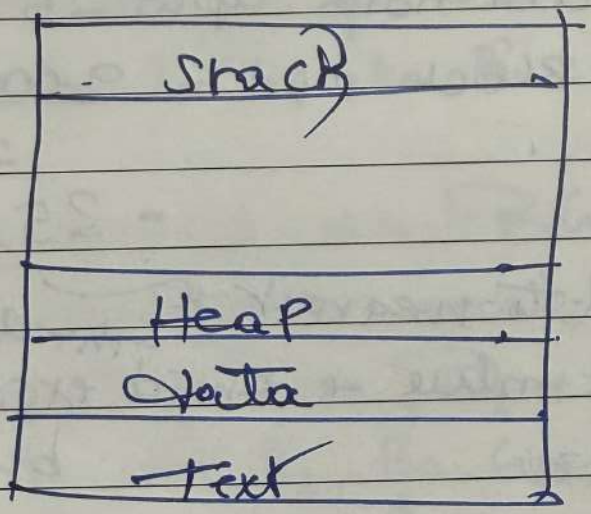
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Architecture of Process:



Text:

It is the actual data transferred from disk to memory to be executed.

Data:

Contains the static as well as Global data.

Heap:

Dynamic Memory Allocation.

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Stack:

Manages function execution & function related data

Stack overflow:

When the function is called, a Stack frame is created for that func. to be stored in the stack.

In a condition like Infinite Recursion, Function be called again & again

⇓

More stack frame created

⇓

Stack gets bigger & reaches a limit

⇓

Stack Overflow.

||

Could be avoided by writing a Base Condition.

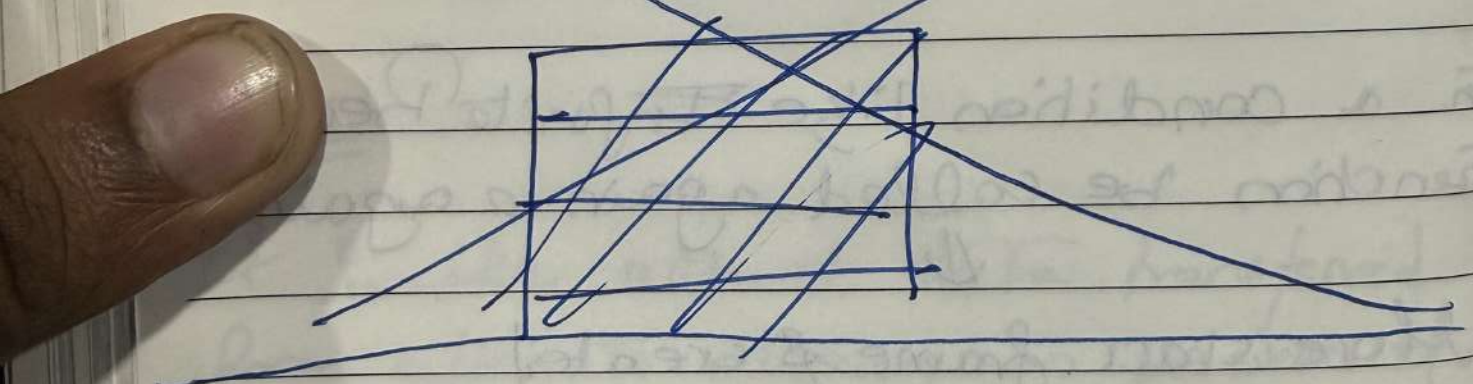
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Process Control Blocks:

How does OS differentiate one process from another? It uses the Attributes of the process.

~~These Attributes are stored in the first block of the process Table, called the Process Control Block~~

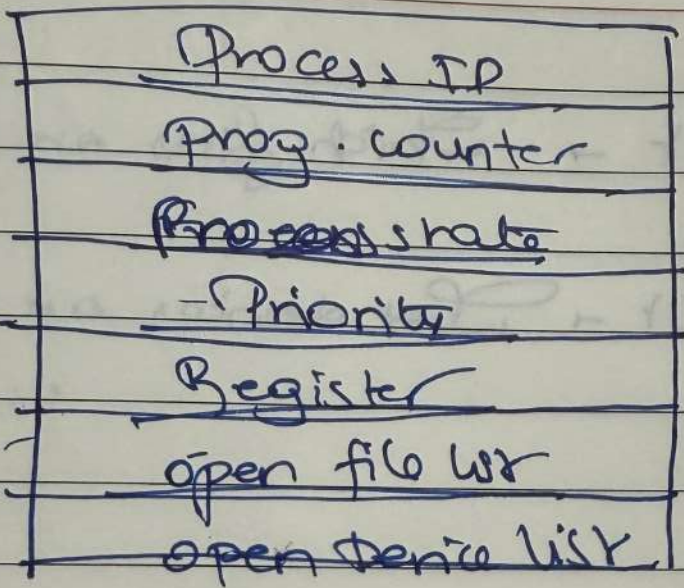


These Attributes are stored in the Process Control Block (PCB). Every process has its own PCB.

~~The structure of the PCB would be:~~

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Process ID → Unique identifier

Program Counter → What Instruction is currently being executed
↓
Keep many it.

Process state → What state process currently in?

Priority → Needs to schedule tasks based on high to low Priority.

Registers → Many Registers like Stack Pointer (SP), Base pointer (BP), Control Register (CR).
↓
small
Temp storage.

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open file list → Which files are open

Open device list → What devices are open.